

Task 8 — Chemistry: Test 4, Organic Molecules

Course: ATAR Chemistry Year 12 (ATCHE), Kent Street SHS, 2026 **Sources:** KSSHS ATCHE Course Outline 2026.docx (Weeks 18–23) and Pearson - Chemistry - Year 12 Unit 4.pdf (Chapter 13 *Structure of organic molecules*, Chapter 14 *Reactions of organic molecules*)

1. What the outline says about Task 8

Field	Detail
Task	Task 8: Test 4 — Organic Molecules
When	Week 23 (Term 3, Week 3)
Immediate content	Pearson 13.6; Lucarelli Ch 13 pp. 140–142, Set 19 pp. 142–144

The test comes at the end of the Organic Molecules sequence, so it covers everything taught in **Weeks 18–23**. Here is exactly what the Resources column lists for each of those weeks:

Weeks	Syllabus content	Pearson	Lucarelli
18–19 (T2 W9–10)	hydrocarbon skeletons and functional groups (alkenes, alcohols, aldehydes, ketones, carboxylic acids, esters, amines, amides); structural formulae, condensed or showing bonds	13.1–13.5 and 14.1–14.3	Ch 10 pp. 108–113 (Set 15 pp. 113–117); Ch 11 pp. 118–127 (Set 16 pp. 128–131)
20–21 (T2 W11, T3 W1)	characteristic reactions used to identify functional groups — addition reactions of alkenes, redox reactions of alcohols, acid–base reactions of carboxylic acids; IUPAC nomenclature up to 8 carbons with simple branching and one functional group; chain, position and cis–trans isomerism	(13.5, 14.1–14.3)	Ch 12 pp. 132–133 (Set 17 pp. 133–135)
22 (T3 W2)	complete combustion of alcohols; oxidation with acidified MnO_4^- or $\text{Cr}_2\text{O}_7^{2-}$ ($1^\circ \rightarrow$ aldehyde $\rightarrow$ carboxylic acid, $2^\circ \rightarrow$ ketone) with characteristic observations and equations; condensation of alcohols with carboxylic acids to form esters; physical properties (boiling point, solubility in water and organic solvents) explained by dispersion forces, dipole–dipole interactions and hydrogen bonds	(14.2, 14.3, 13.3)	Ch 12 pp. 135–137 (Set 18 pp. 138–139)
23 (T3 W3)	empirical and molecular formulae determined by calculation, and structure established from the chemical reactions the compound undergoes plus other analytical data	13.6	Ch 13 pp. 140–142 (Set 19 pp. 142–144)

So the examinable scope is Pearson 13.1 → 13.6 and 14.1 → 14.3. (Pearson 14.4 organic synthesis, and chapters 15–18 on polymers, soaps, proteins and enzymes, belong to Task 9 and later.)

Two notes on the source material. The **Lucarelli** workbook is not in the repo, so page references to it are carried over from the outline but its questions aren't reproduced here. The **Pearson** PDF is a scan with no text layer, so it was read by OCR — section numbering and the summary points below are taken from the book itself. Separately, [Experiment 31.pdf](#) and [Exp 32 - Esters.pdf](#) belong to **Task 7** (Week 22 Practical Validation — Reactions of Organic Molecules), not this test; there's a companion guide for those in the repo.

2. Pearson 13.1 — Diversity of carbon compounds

Carbon forms more compounds than all other elements combined because it has **four valence electrons** (so it forms four strong covalent bonds), it bonds readily to itself in chains, branches and rings, and it forms stable multiple bonds.

Homologous series — a family in which each member has one more **-CH₂-** unit than the last. Members share a **general formula**, similar chemical properties, and show a gradual trend in physical properties.

Series	General formula	Bonding	Saturated?
Alkanes	C_nH_{2n+2}	C-C single bonds only	saturated
Alkenes	C_nH_{2n}	contains one C=C double bond	unsaturated

Representing molecules

- **Molecular formula** — C₄H₁₀. Says what atoms, not how they're joined.
- **Structural formula** — every atom and bond drawn. Bonds are usually drawn at right angles for clarity, even though the real geometry around each carbon is **tetrahedral** (~109.5°), which is why carbon chains are actually zig-zagged, not straight.
- **Condensed structural formula** — CH₃CH₂CH₂CH₃, or CH₃CH(CH₃)CH₂CH₃ where branches sit in brackets.

Naming an alkene (the flowchart in 13.1): identify the longest unbranched carbon chain containing the C=C → number the chain so the **double bond gets the lowest possible number** → name and number any branches → check whether cis-trans isomerism exists and add the prefix if so. Example: CH₃CH=CHCH₂CH₃ is pent-2-ene.

3. Pearson 13.2 — Functional groups

A **functional group** is the group of atoms or bonds responsible for a molecule's characteristic chemical properties. Most organic compounds can be regarded as hydrocarbons in which one or more hydrogens has been replaced.

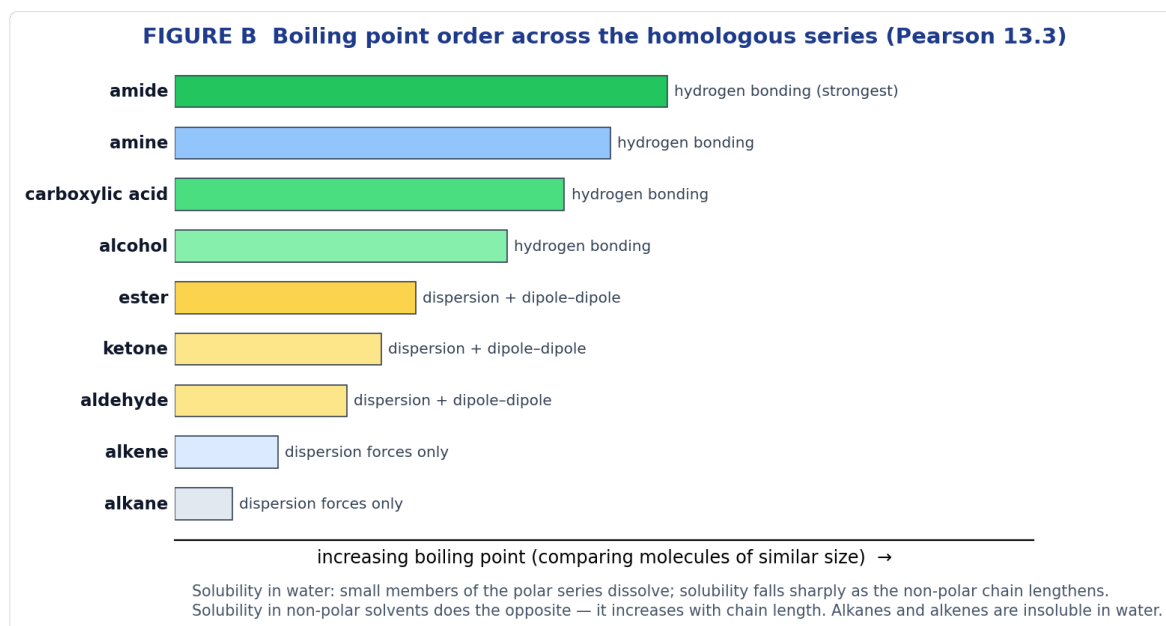
Class	Functional group	Where it sits	Naming	Example (condensed)	Example name
Alkene	C=C	anywhere in the chain	suffix -ene + locant	CH ₃ CH=CH ₂	propene
Alcohol	-OH hydroxyl	anywhere	drop -e, add -ol + locant	CH ₃ CH ₂ OH	ethanol
Aldehyde	-CHO carbonyl at the end	end only	drop -e, add -al (no locant needed)	CH ₃ CHO	ethanal
Ketone	-CO- carbonyl within the chain	not the end	drop -e, add -one + locant	CH ₃ COCH ₃	propanone
Carboxylic acid	-COOH carboxyl	end only	add -oic acid	CH ₃ COOH	ethanoic acid
Ester	-COO-	within the chain	two words: alkyl (from the alcohol) + alkanoate (from the acid)	CH ₃ COOCH ₂ CH ₃	ethyl ethanoate
Amine (primary)	-NH₂ amino	anywhere	drop -e, add -amine + locant	CH ₃ CH ₂ NH ₂	ethanamine
Amide (primary)	-CONH₂	end only	drop -e, add -amide	CH ₃ CONH ₂	ethanamide

The **carbonyl group, C=O**, is **shared** by aldehydes, ketones, carboxylic acids, esters and amides — what distinguishes them is what else is attached to that carbon and **where in the chain it sits**.

Why some groups can't have position isomers: the carboxyl, aldehyde and amide groups can only occur at the **end** of a chain, because that carbon already uses a double bond to one oxygen plus its other bonds. So there is no "propan-2-oic acid". Alcohols, amines, ketones and esters *can* vary in position and therefore have **positional isomers**.

Ester naming, worked: for CH₃CH₂COOCH₃ — the part on the right of the -COO- came from the **alcohol** (CH₃- → methyl), the part on the left came from the **acid** (CH₃CH₂COOH, propanoic acid → propanoate). Name: **methyl propanoate**.

4. Pearson 13.3 — Physical properties



Boiling point is determined by intermolecular forces. Dispersion forces are **always** present; on top of those a molecule may have **dipole-dipole** attractions (polar C=O) or **hydrogen bonds** (O-H or N-H).

- Alkanes and alkenes are non-polar → **dispersion forces only** → lowest boiling points. Strength grows with molecular size and with a less-branched shape (more surface contact).
- Aldehydes, ketones and esters have a polar carbonyl but no O-H → **dispersion + dipole-dipole** → intermediate.
- Alcohols, carboxylic acids, amines and amides can **hydrogen bond** → highest.
- Comparing molecules of **similar size**, Pearson gives the order: **alkanes < alkenes < aldehydes < ketones < esters < alcohols < carboxylic acids < amines < amides**.

Solubility in water depends on whether water molecules can hydrogen bond to the solute, so it is a tug-of-war between the polar functional group and the non-polar hydrocarbon "tail":

- Alkanes and alkenes are **insoluble** in water; they *are* soluble in non-polar organic solvents.
- The **small** members of the alcohol, aldehyde, ketone, ester, amine and amide series are generally soluble in water, and solubility **falls rapidly as chain length increases**.
- Solubility in **non-polar** solvents does the opposite — it **increases** with chain length.

Answer structure that scores: name the forces present in each substance → compare their relative strength → link to the observation. e.g. "*Butan-1-ol can hydrogen bond through its -OH group, whereas butanal has only dipole-dipole attractions and dispersion forces between its molecules. More energy is required to separate the alcohol molecules, so butan-1-ol has the higher boiling point.*"

5. Pearson 13.4 — Isomers

Isomers contain the same number and type of atoms arranged differently, so they have the **same molecular formula but different structures** — and therefore different properties.

Type	What differs	Example
Chain (skeletal) isomers	the branching of the carbon skeleton	butane and 2-methylpropane (C_4H_{10})
Positional isomers	the position of a functional group on the chain	butan-1-ol and butan-2-ol ($C_4H_{10}O$); pent-1-ene and pent-2-ene
Geometric (cis-trans) isomers	arrangement in space , atoms joined in the same order	cis- and trans-but-2-ene

cis-trans isomerism requires restricted rotation — which occurs about a **C=C double bond** (or in a ring) — **and** two *different* groups attached to the carbon at **each** end of the double bond. If either double-bonded carbon carries two identical groups (e.g. two H atoms), there are **no** cis-trans isomers.

- **cis** — the two like/priority groups are on the **same** side of the double bond.
- **trans** — on **opposite** sides.

The shortest alkane with chain isomers is **butane** ($C_4H_{10} \rightarrow 2$ isomers).

6. Pearson 13.5 — IUPAC nomenclature

The naming algorithm, in order:

1. **Find the longest unbranched carbon chain** — that gives the parent name (meth-, eth-, prop-, but-, pent-, hex-, hept-, oct-). The chain may bend around corners in the drawing; count carefully.
2. **Identify the functional group(s)** and decide which is named as a **suffix** (the highest priority one) and which become **prefixes**.
3. **Number the chain** so the carbon attached to the **highest-priority functional group gets the lowest number**. (Where there's no functional group to fix it, number so the substituents get the lowest locants.)
4. **Insert the locant before each group**, use **di-, tri-, tetra-** multipliers for repeats, and list prefixes **alphabetically**.
5. Separate numbers from numbers with commas and numbers from letters with hyphens.

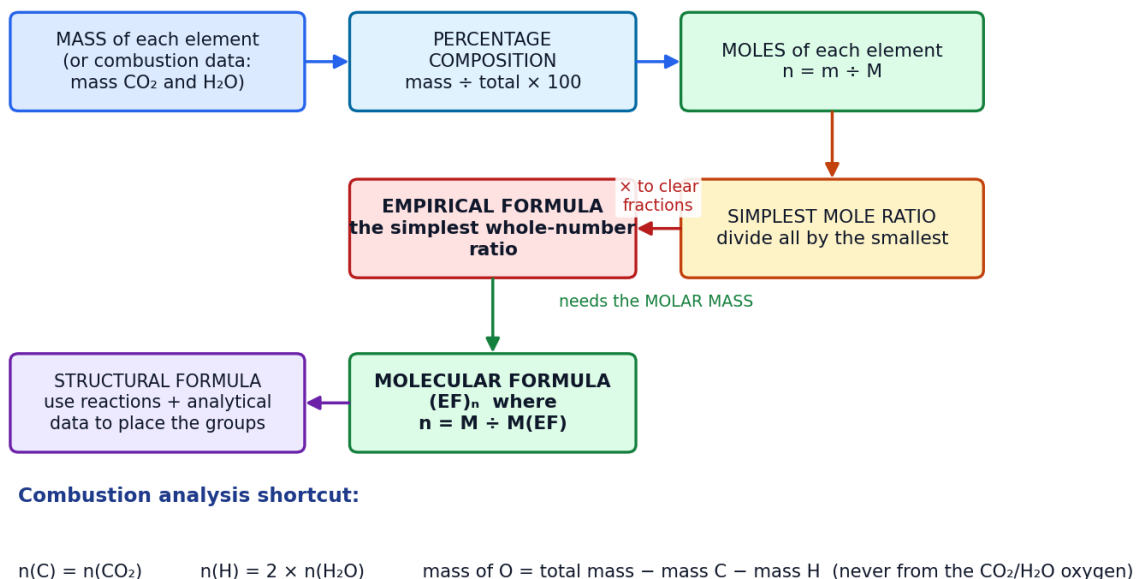
Priority order for the suffix (highest first): carboxylic acid > ester > amide > aldehyde > ketone > alcohol > amine > alkene. Lower-priority groups become prefixes (hydroxy-, amino-, oxo-, alkyl-).

Fixing wrong names — the classic exam item

Given (wrong)	Why it's wrong	Correct
<i>3-aminopropane</i>	propane has only 3 carbons; numbering must give the amino group the lowest locant, and C3 from one end is C1 from the other. The group should also be the suffix.	propan-1-amine (1-aminopropane)
<i>3-ethyl-hydroxy-2-butane</i>	the hydroxyl has no locant and isn't expressed as the -ol suffix; and the parent chain chosen isn't the longest — including the ethyl branch gives a 5-carbon chain	3-methylpentan-2-ol

7. Pearson 13.6 — Determining formulae (the Week 23 content)

FIGURE C Determining formulae of organic compounds (Pearson 13.6)



Percentage composition of each element:

$$\%_{\text{element}} = (\text{mass of the element present} \div \text{total mass of the compound}) \times 100$$

Worked example 1 — percentage composition. 8.38 g of a compound contains 5.44 g carbon, 1.13 g hydrogen and 1.81 g oxygen.

$$\begin{aligned} \% \text{C} &= 5.44 / 8.38 \times 100 = 64.9 \% \\ \% \text{H} &= 1.13 / 8.38 \times 100 = 13.5 \% \\ \% \text{O} &= 1.81 / 8.38 \times 100 = 21.6 \% \end{aligned} \quad (\text{check: they sum to } 100 \%)$$

Worked example 2 — empirical formula from percentages. A compound is 52.2 % carbon, 13.0 % hydrogen, remainder oxygen.

Assume 100 g, so the masses in grams equal the percentages. $\text{O} = 100 - 52.2 - 13.0 = 34.8 \%$

$$\begin{aligned} n(\text{C}) &= 52.2 / 12.01 = 4.346 \text{ mol} \\ n(\text{H}) &= 13.0 / 1.008 = 12.90 \text{ mol} \\ n(\text{O}) &= 34.8 / 16.00 = 2.175 \text{ mol} \end{aligned}$$

Divide through by the smallest (2.175):

$$\text{C} : \text{H} : \text{O} = 1.998 : 5.93 : 1.000 \approx 2 : 6 : 1$$

Empirical formula = $\text{C}_2\text{H}_6\text{O}$

Worked example 3 — molecular formula from empirical formula and molar mass. A sample of 0.8390 mol has a mass of 152.7 g.

$$M = m / n = 152.7 / 0.8390 = 182.0 \text{ g mol}^{-1}$$

If the empirical formula were CH_2O ($M = 30.03$): $n = 182.0 / 30.03 = 6.06$ $\times$ not a whole number

With the empirical formula $\text{C}_3\text{H}_7\text{O}_3$ ($M = 91.09$): $n = 182.0 / 91.09 = 2.00$ $\checkmark$

Molecular formula = $(\text{C}_3\text{H}_7\text{O}_3)_2 = \text{C}_6\text{H}_{14}\text{O}_6$

*The check matters: **n must come out as a whole number.*** If it doesn't, either the empirical formula or the molar mass is wrong — say so in your answer.

Worked example 4 — combustion analysis. 0.500 g of a compound containing only C, H and O burns completely to give 0.955 g CO₂ and 0.587 g H₂O.

$$\begin{aligned}n(\text{CO}_2) &= 0.955 / 44.01 = 0.02170 \text{ mol} \rightarrow n(\text{C}) = 0.02170 \text{ mol} \rightarrow m(\text{C}) = 0.02170 \times 12.01 = 0.2606 \text{ g} \\n(\text{H}_2\text{O}) &= 0.587 / 18.02 = 0.03258 \text{ mol} \rightarrow n(\text{H}) = 2 \times 0.03258 = 0.06515 \text{ mol} \rightarrow m(\text{H}) = 0.0657 \text{ g}\end{aligned}$$

$$m(\text{O}) = 0.500 - 0.2606 - 0.0657 = 0.1737 \text{ g} \rightarrow n(\text{O}) = 0.1737 / 16.00 = 0.01086 \text{ mol}$$

$$\text{Divide by the smallest (0.01086):} \quad \text{C} : \text{H} : \text{O} = 2.00 : 6.00 : 1.00$$

Empirical formula = C₂H₆O. If M = 46.07 g mol⁻¹ then n = 1, so the molecular formula is also C₂H₆O.

The trap: get the oxygen by **subtracting** from the total sample mass. Never take it from the oxygen in the CO₂ or H₂O — that oxygen came from the O₂ used to burn the sample.

Establishing the structure (the second half of the Week 23 dot point): once you have the molecular formula, use the **reactions** the compound undergoes to place the functional group. C₂H₆O could be ethanol or methoxymethane; if it is oxidised by acidified dichromate to an acid, and reacts with sodium to give H₂, it is **ethanol**.

8. Pearson 14.1 — Reactions of alkenes

Combustion — alkenes burn in oxygen to give carbon dioxide and water: $\text{C}_3\text{H}_6(\text{g}) + 4\frac{1}{2}\text{O}_2(\text{g}) \rightarrow 3\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{g})$

Addition reactions — the C=C double bond opens and one new group joins each carbon, converting an unsaturated molecule into a **saturated** one:

Reagent	Conditions	Product	Example
H₂	Ni (or Pt) catalyst	alkane	$\text{C}_2\text{H}_4 + \text{H}_2 \rightarrow \text{C}_2\text{H}_6$
X₂ (Br ₂ , Cl ₂)	room temperature	dihaloalkane	$\text{C}_2\text{H}_4 + \text{Br}_2 \rightarrow \text{CH}_2\text{BrCH}_2\text{Br}$ (1,2-dibromoethane)
HX (HCl, HBr)	—	haloalkane	$\text{C}_2\text{H}_4 + \text{HCl} \rightarrow \text{CH}_3\text{CH}_2\text{Cl}$ (chloroethane)
H₂O	H ₂ SO ₄ (or H ₃ PO ₄) catalyst	alcohol (hydration)	$\text{C}_2\text{H}_4 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CH}_2\text{OH}$

Two possible products: when the alkene is not symmetrical, the new groups can add either way round. But-1-ene + HCl gives both 1-chlorobutane and 2-chlorobutane. Show both, and note which forms in greater amount if your course requires it.

Addition polymerisation — alkenes add to each other, the double bonds opening to form long chains containing **only C-C single bonds**. Ethene → polyethene. The repeating unit has the same atoms as the monomer, so you can work backwards from a polymer's repeat unit to its monomer by putting the double bond back.

Test for unsaturation: bromine water is **decolourised** (orange → colourless) by an alkene, because Br₂ is consumed in an addition reaction. An alkane leaves it orange.

9. Pearson 14.2 — Reactions of alcohols

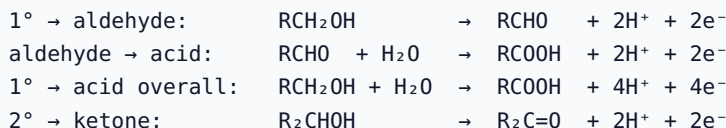
Combustion — all alcohols burn completely in excess oxygen to CO_2 and H_2O , strongly exothermically, which is why ethanol is used as a fuel (E10, methylated spirits, bioethanol from fermented sugar cane): $\text{C}_2\text{H}_5\text{OH}(\text{l}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{g})$

Oxidation — with strong inorganic oxidising agents: acidified **dichromate** ($\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$) or acidified **permanganate** (KMnO_4/H^+). What happens depends on the **class** of alcohol, which is why this is used as an identification test.

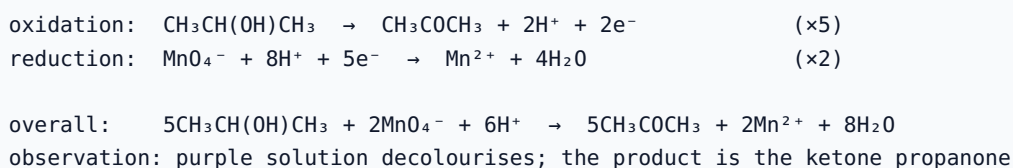
Class	-OH is on a carbon bonded to	Product	Observation
Primary (1°)	1 alkyl group	aldehyde , then on further heating the carboxylic acid	colour change occurs
Secondary (2°)	2 alkyl groups	ketone (no further oxidation)	colour change occurs
Tertiary (3°)	3 alkyl groups	no reaction	no colour change

- **Dichromate:** orange $\text{Cr}_2\text{O}_7^{2-} \rightarrow$ green Cr^{3+} .
- **Permanganate:** purple $\text{MnO}_4^- \rightarrow$ colourless Mn^{2+} .
- To stop at the **aldehyde**, use milder conditions — lower temperature, shorter reaction time — and **distil the aldehyde off** as it forms (its boiling point is lower, since it can't hydrogen bond). To go all the way to the **carboxylic acid**, heat under **reflux** with excess oxidant.
- A tertiary alcohol resists oxidation because there is **no hydrogen on the carbon carrying the -OH**, so the $\text{C}=\text{O}$ cannot form without breaking a C-C bond.

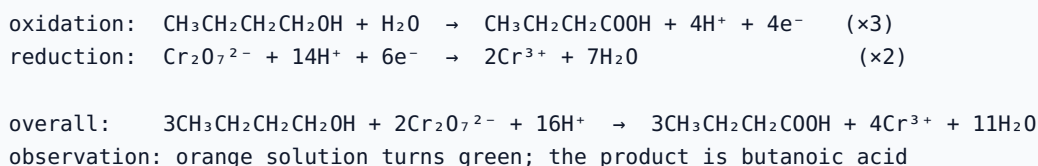
Oxidation half equations



Worked redox equation A — propan-2-ol (2°) heated with acidified permanganate



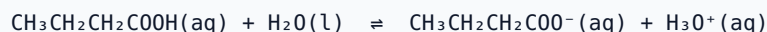
Worked redox equation B — butan-1-ol (1°) heated with acidified dichromate



Always check charge as well as atoms. In equation B: left $-4 + 16 = +12$; right $4 \times (+3) = +12$. ✓

10. Pearson 14.3 — Reactions of carboxylic acids

They are weak acids — only partially ionised in water, so the equilibrium lies to the left:

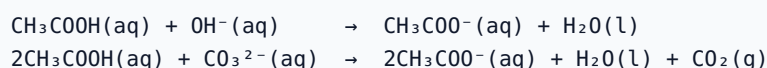


In such a solution, species run from most to least abundant: **H₂O** > **un-ionised acid molecules** > **carboxylate ions** $\approx$ **H₃O⁺** > **OH⁻** — because the acid is weak, most molecules stay intact. The **conjugate base** of propanoic acid is the **propanoate ion**, CH₃CH₂COO⁻.

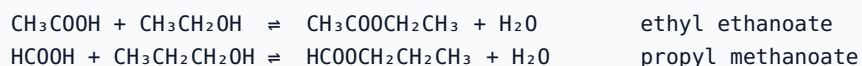
Acid-base reactions (these identify a carboxylic acid):

With	Products	Example
reactive metal	salt + hydrogen gas	$2\text{CH}_3\text{COOH} + \text{Mg} \rightarrow \text{Mg}(\text{CH}_3\text{COO})_2 + \text{H}_2$
metal oxide	salt + water	$2\text{CH}_3\text{COOH} + \text{MgO} \rightarrow \text{Mg}(\text{CH}_3\text{COO})_2 + \text{H}_2\text{O}$
metal hydroxide	salt + water	$\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$
carbonate / hydrogencarbonate	salt + water + carbon dioxide	$2\text{CH}_3\text{COOH} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{CH}_3\text{COONa} + \text{H}_2\text{O} + \text{CO}_2$

Ionic equations — leave the weak acid **un-ionised**, because it mostly isn't ionised:



Esterification (condensation) — carboxylic acid + alcohol, with a small amount of concentrated H₂SO₄ as catalyst, gives an **ester + water**. It is the **-OH from the acid** and the **-H from the alcohol** that combine to form the water:



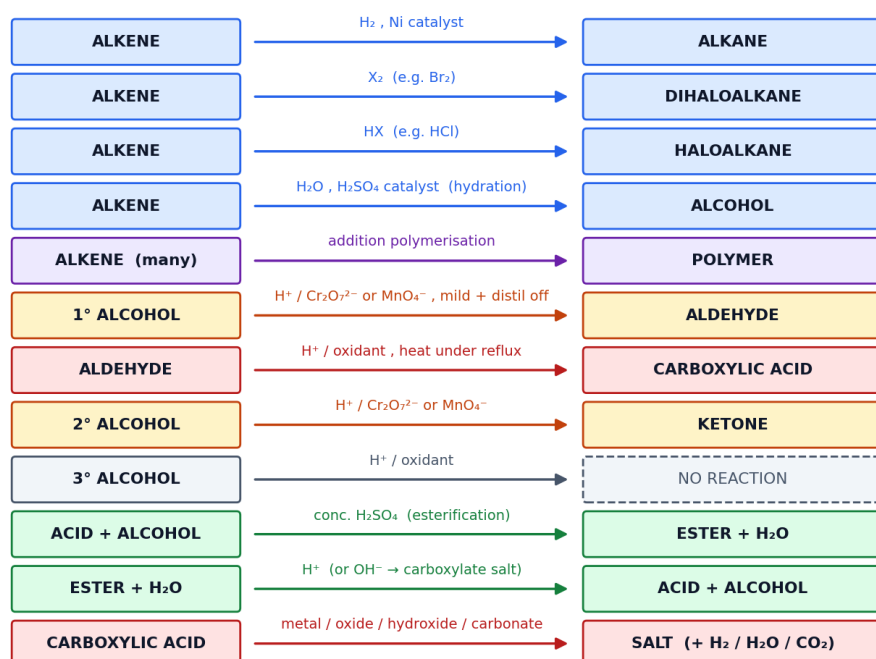
Hydrolysis of esters — the reverse reaction. Because esterification is reversible, an ester can be split:

- with **water and an acid catalyst** → the carboxylic **acid** + the alcohol;
- with an **alkali** (e.g. NaOH) → the **carboxylate salt** + the alcohol. This one goes essentially to completion, because the base removes the acid product.

11. Reaction summary and identification tests

FIGURE A Reaction summary for Task 8 (Pearson 14.1-14.3)

Learn the reagents on the arrows — that is what identifies a functional group. Colour changes: $\text{Cr}_2\text{O}_7^{2-}$ orange $\rightarrow$ green; MnO_4^- purple $\rightarrow$ colourless.



Identifying an unknown compound — the syllabus dot point says reactions "can be used to identify the functional group present", so expect a question like "describe a test to distinguish X from Y".

Test	Positive result	Identifies
Add bromine water	orange $\rightarrow$ colourless	alkene (addition, unsaturation)
Add acidified $\text{Cr}_2\text{O}_7^{2-}$ and warm	orange $\rightarrow$ green	1° or 2° alcohol (or an aldehyde)
Same test, no colour change	stays orange	3° alcohol , alkane, ketone, ester
Add Na_2CO_3 or NaHCO_3	effervescence (CO_2)	carboxylic acid
Test with litmus / pH	blue litmus $\rightarrow$ red, pH < 7	carboxylic acid
Add sodium metal	steady bubbles of H_2	alcohol (or carboxylic acid, which also gives CO_2 with carbonate)
Smell and solubility	sweet fruity smell, insoluble layer, neutral	ester

To separate a **primary** from a **secondary** alcohol: oxidise fully and test the product. A primary alcohol ends up as a **carboxylic acid**, which will effervesce with carbonate and redden litmus; a secondary alcohol gives a **ketone**, which does neither.

12. Exam technique and the traps in this topic

- **Name the reagent *and* the condition.** "Acidified dichromate, heated under reflux" earns the mark; "an oxidising agent" often doesn't.
 - **Quote colour changes with the species:** orange $\text{Cr}_2\text{O}_7^{2-}$ → green Cr^{3+} ; purple MnO_4^- → colourless Mn^{2+} .
 - **Balance redox equations methodically:** atoms other than O and H first → H_2O for oxygen → H^+ for hydrogen → electrons for charge → scale and add → cancel $\text{H}^+/\text{H}_2\text{O}$. Check **charge** at the end.
 - **Esters are named backwards** — alkyl from the alcohol first, alkanoate from the acid second. *Ethyl butanoate* and *butyl ethanoate* are different compounds.
 - **Don't ionise a weak acid** in an ionic equation.
 - **Carboxyl, aldehyde and amide groups sit only at the end of a chain** — so no locant, and no positional isomers for them.
 - **cis-trans needs two different groups at *each* end** of the $\text{C}=\text{C}$. Check both carbons before claiming geometric isomers exist.
 - **In combustion analysis, get oxygen by subtraction.**
 - **Check that n is a whole number** when converting an empirical formula to a molecular formula.
 - **Explain boiling point and solubility with the actual forces**, in a comparison: dispersion (always), dipole-dipole (polar $\text{C}=\text{O}$), hydrogen bonding ($\text{O}-\text{H}$, $\text{N}-\text{H}$) — then compare strengths and conclude.
 - **Longest chain, lowest locants** — most naming errors come from picking a shorter chain that misses a branch, or numbering from the wrong end.
-

13. Checklist

- ☐ State the general formulae C_nH_{2n+2} and C_nH_{2n} and explain what a homologous series is
- ☐ Convert between molecular, structural and condensed structural formulae
- ☐ Recognise and draw all eight functional groups, and know which ones can only sit at the end of a chain
- ☐ Name and draw compounds up to 8 carbons with simple branching and one functional group
- ☐ Fix an incorrectly named compound and say which rule was broken
- ☐ Draw chain, positional and cis-trans isomers, and state the conditions for cis-trans to exist
- ☐ Rank boiling points across homologous series and justify with the specific intermolecular forces
- ☐ Explain water solubility and non-polar-solvent solubility, and the effect of chain length on each
- ☐ Calculate percentage composition, empirical formula and molecular formula
- ☐ Do a combustion analysis, getting oxygen by subtraction
- ☐ Deduce a structure from a molecular formula plus reaction evidence
- ☐ Write equations for alkene combustion and for addition of H_2 , X_2 , HX and H_2O
- ☐ Predict a polymer from a monomer, and a monomer from a polymer repeat unit
- ☐ Write equations for the complete combustion of an alcohol
- ☐ Classify an alcohol and predict its oxidation product, with the colour change
- ☐ Write oxidation and reduction half equations and combine them into a balanced redox equation
- ☐ Explain why a tertiary alcohol is not oxidised
- ☐ Write the ionisation equation for a weak carboxylic acid and rank the species present
- ☐ Write equations for a carboxylic acid with metals, oxides, hydroxides and carbonates, molecular and ionic
- ☐ Write esterification equations, name the ester, and describe acid and alkaline hydrolysis
- ☐ Describe a chemical test to distinguish any two of: alkene, $1^\circ/2^\circ/3^\circ$ alcohol, aldehyde, ketone, carboxylic acid, ester